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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

II Semester B. E. Regular / Supplementary Examinations Aug-2024

COMMON TO AS /CH / IM / ME

CHEMISTRY OF ENGINEERING MATERIALS

Time: 03 Hours *Maximum Marks: 100*

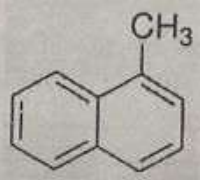
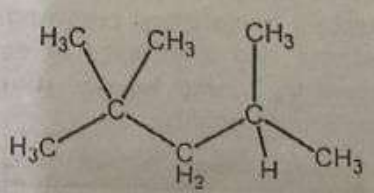
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 & 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10, and 11 lab components (compulsory).

PART-A

		M	BT	CO
1.1	Why *HCV* is always higher than *LCV*?	01	3	3
1.2	Out of the two additives ethyl nitrate and ethanol identify which can improve the cetane number.	01	3	2
1.3	Identify the battery characteristic used to predict the life span of secondary battery.	01	2	3
1.4	Write the anodic reaction of aqueous lithium air battery.	01	3	2
1.5	Identify a metal from the given list which is protected by a layer of its own oxide. List of metals are *Cu, Au, Fe* and *Al*	01	3	1
1.6	Justify why Iron does not rust when exposed to dry air.	01	3	3
1.7	After the functionalization identify the hybridization of carbon atom at the functionalized site.	01	2	3
1.8	Justify gold nanoparticles exhibit different colors.	01	4	3
1.9	Give reason why glass transition temperature of polypropylene is lower compared to polyvinylchloride.	01	2	2
1.10	What is the commercial name of Phenol-formaldehyde resin?	01	2	1

PART B

a	What is knocking in petrol IC engine and diesel engine? By applying the knowledge of chemistry, Identify and highlight the significance of the below hydrocarbon in evaluation of knocking characteristics of petrol/diesel fuel. Mention one adverse of knocking.  α-Methyl-Naphthalene  Iso-octane	07	2	2
b	What is the significance of green hydrogen? Discuss the production and storage of green hydrogen.	07	2	2

3	a	Outline the construction and working of the rechargeable $LiCoO_2$ battery with suitable diagram. With the help of reversible redox reactions, discuss the $Li+$ ion exchange mechanism between anode and cathode.	07	2	2
	b	With neatly labeled diagram explain Methanol-Oxygen fuel cell and the mechanism of conversion of chemical energy into electrical energy using anodic and cathodic reactions taking place in the presence of H_2SO_4 electrolyte. Discuss any one application of fuel cell.	07	2	2
		OR			
4	a	Outline the fabrication and working of the lithium-air battery with a suitable diagram and reactions. Discuss its limitation.	07	2	2
	b	With neat labeled diagram explain the construction and working of Quantum dot sensitized solar cell.	07	2	2
5	a	How rate of corrosion is affected by the following factors i) Corrosion product ii) Relative anodic and cathodic area iii) Temperature	07	3	3
	b	What are the advantages of electroless plating over electroplating? Explain the process of electroplating of chromium for engineering applications and indicate the reasons for not employing chromium as anode.	07	2	2
		OR			
6	a	What is anodic metallic coating? Justify the need of galvanized metal sheets used for outdoor purpose. Explain the process of galvanization.	07	3	3
	b	Discuss with neat labeled diagram the corrosion of iron rod exposed in environment based on electrochemical theory.	07	2	2
7	a	Discuss the reactions and steps involved with neat labeled diagram in the preparation of nano materials by sol-gel method.	07	2	1
	b	Illustrate the synthesis of grapheme by Hummer's method. Comment on its mechanical and optical properties. Mention any one application of grapheme.	07	2	1
		OR			
	a	Discuss the principle and procedure in the preparation of nano materials by solution combustion synthesis.	07	2	1
	b	Illustrate the synthesis of carbon nanotube by modified CVD method with neat labeled diagram. Mention any two application of carbon nanotube.	07	2	1
	a	What is biodegradable polymer? Explain the synthesis of poly lactic acid with application.	07	2	2
	b	Discuss the synthesis of glass fiber. How it can be used in polymer composites and list two applications.	07	2	2
		OR			

10	a	What is glass transition temperature of a polymer? How the following factors affect glass transition of polymer i) Chemical Structure ii) Stereo regularity iii) Flexibility	07	2	2
	b	Discuss the synthesis of carbon fiber. How it can be used in polymer composites and list two applications?	07	2	2
		LAB COMPONENT			
11	a	Discuss the procedure for conductometric titration of strong acid with strong base procedure. Plot the model graph and explain the nature of graph.	10	2	3
	b	Explain the colorimetry principle, procedure and calculation involved in the estimation of copper from copper sulphate solution.	10	2	3

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COMMON TO AS/CHIM/ME

CHEMISTRY OF ENGINEERING MATERIALS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first two pages of the answer book only.
2. Answer SIX full questions from Part B. In Part B question number 2 and 11 are compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.

PART A

1	1.1	Give an example for lead free anti knocking agent used along with petrol.	01
	1.2	List any two limitations of biodiesel.	01
	1.3	What is the main role of solid electrolyte interphase in Lithium-ion battery?	01
	1.4	Acid electrolyte is preferred over alkali electrolyte in direct methanol fuel cell. Justify.	01
	1.5	Identify the possible type of corrosion when an iron rod is tied with copper wire in moist aerated environment.	01
	1.6	Chromium anode is not preferred in the electroplating of chromium in hard chrome plating. Justify.	01
	1.7	Name the precursor used in the solution combustion synthesis of metal nano oxide.	01
	1.8	Preparation of carbon nanotubes is preferred in inert atmosphere . Justify.	01
	1.9	Name the factor which is inversely proportional to glass transitional temperature of the polymer.	01
	1.10	Mention any one application of poly ether sulfone.	01

PART B

2	a	What is knocking in petrol engine? Explain the factors affecting knocking in petrol engine.	07
	b	List the types and characteristics of advanced rocket propellants.	07
3	a	Outline the construction and working of lithium cobalt oxide battery along with neat labeled diagram and chemical reactions involved in it.	07
	b	Explain the fabrication and working of Quantum Dot sensitized solar cell with neat labeled diagram.	07

OR

4	a	Illustrate the construction and working of electrical double layer capacitor and pseudo capacitor along with the materials used for such type of super capacitors.	07
	b	What are fuel cells? Outline the construction and working of direct methanol fuel cell along with its applications.	07
5	a	Explain the corrosion behavior of iron beam placed in the metro construction area with the help of electrochemical theory.	07
	b	Illustrate the process of galvanization of iron and anodization of aluminum along with chemical reactions and neat labeled diagram.	07
		OR	
6	a	Outline any three factors affecting the rate of corrosion when a metal is under moist aerated environment.	07
	b	Explain the electro less plating of copper with bath conditions and chemical reactions. Mention any one application of electroless plating.	07
7	a	Explain the following properties of nano materials. i. optical property ii. catalytic property iii. Surface area	07
	b	Outline the preparation of carbon nanotube by modified CVD technique with neat labeled diagram.	07
		OR	
8	a	Outline the synthesis of metal nano oxide by solgel technique and explain the processing of gel along with the chemical reactions. Name any one nano material prepared in this method.	07
	b	Explain the synthesis of graphene by modified hummer method and mention the industrial importance of graphene.	07
9	a	What is glass transition temperature (Tg) of the polymer? Explain any two factors affecting the Tg of the polymers with an example each.	07
	b	Outline the synthesis of carbon fiber along with the steps involved in it.	07
		OR	
10	a	Explain the synthesis of following thermoplastic polymers with chemical reactions involved in it. i. Poly carbonate ii. Poly ether sulfone.	07
	b	Outline the process involved in the synthesis of glass fiber, explain the procedure of fiber drawing. Mention any two industrial applications of glass fiber composites.	07
		Lab Component	
11	a	Explain the experimental procedure involved in the determination of estimation of iron in hematite ore using redox principle by using potassium dichromate along with calculations.	10
	b	Outline the estimation of amount of sodium in the given solution using flame photometry. Explain the nature of calibration plot and calculation involved in such determination.	10